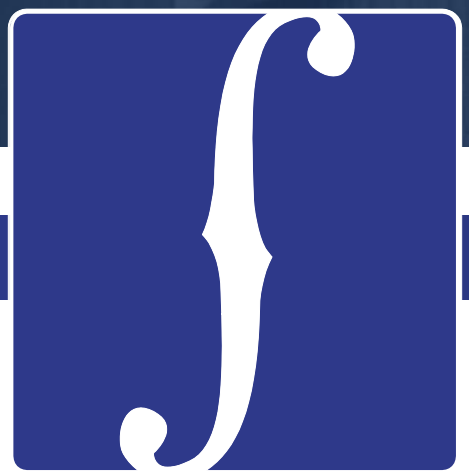




CELLO

## Agency Prepayment Model: Curtailment

February 2016

## Outline

---

- Introduction to Curtailment
- Description of Data
- Empirical Observations and Comparisons
- Description of Framework
- Results
  - Spline Results
  - Final Fixed-rate 30-Year Curtailment Vector

## Introduction to Curtailment

---

- Standard Curtailment Definition:
  - Principal reduction that is greater than the standard amortization schedule for that period
  - Does not include Defaults, Turnover or Refinances
- Identifying Curtailments:
  - Curtailments are not explicitly identified in any data source (eMBS Loan Level data or GSE Loan Level data)
  - Other prepayment types (voluntary vs involuntary) are identified with a flag (GSE LL data) but not in eMBS data
- Research Curtailment Definition:
  - Use Standard Definition while excluding anything where the balance is reduced to zero
  - This has the drawback that we are excluding full payoffs
  - It has the benefit of separating curtailments from Defaults, Turnover and Refinances (as they reduce the balance to 0)
- Empirical Formulas for identifying Curtailments:
  - GSE:  $\text{abs}(\text{balance difference between two months} + \text{scheduled principal pmt}) < 0$
  - eMBS:  $1 - \text{current balance} / \text{scheduled amortization balance}$

## Description of Data

---

- GSE Loan Level data
  - ~ 40 M loans with data as far back as 1999
  - Loan Age up to 180 months
  - Has indicators for Prepayment (voluntary) and Default (involuntary)
- Citi data: Table of data from a 1997 paper by Qiang Fu
  - ~ 25 K loans from Citicorp Mortgage, Inc. (all 30-Year Fixed-rate)
  - Origination years from 1967-1995 and observed curtailments over a period of 21 months
  - This serves as our reference for late maturity curtailments which can be significant
- eMBS Loan Level data
  - ~ 35 M loans with data as far back as 2006
  - Loan Age up to 360 months (sparse data for very seasoned loans)
- eMBS Pool data
  - ~ 800 K pools with data as far back as 1994
  - Loan Age up to 360 months (less sparse than loan level data)

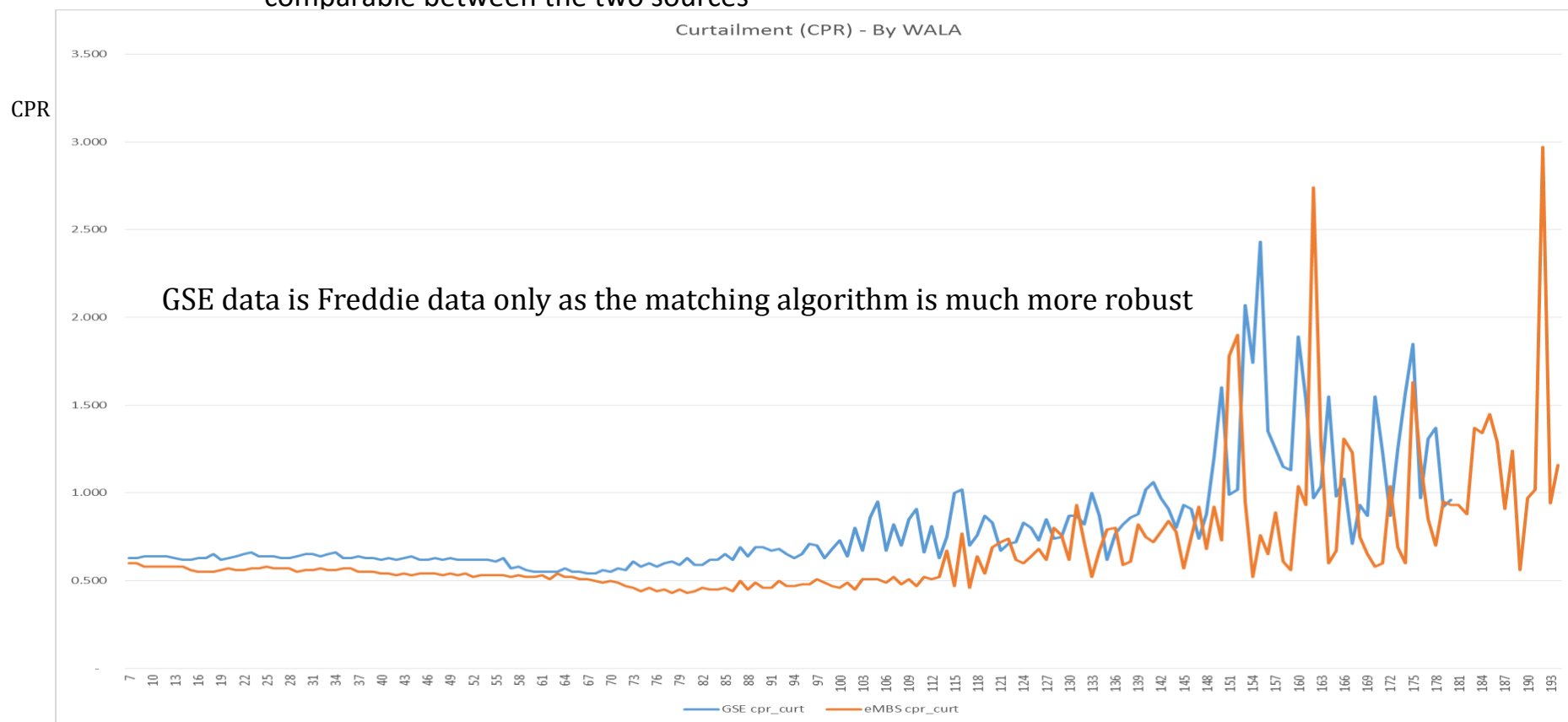
## Empirical Observations (GSE Loan Level Data)

- Curtailment
  - Occurs ~ 9 times in the life of a loan
  - Average Curtailment amount is ~ \$3,100

|                    | Loan Count | Avg Balance | Count Curtailment | Amount Curtailment | Avg # Curtailment | % Bal Curtailment | Avg Curtail Amount |
|--------------------|------------|-------------|-------------------|--------------------|-------------------|-------------------|--------------------|
| <b>Total</b>       | 15,563,076 | 115,336     | 135,717,582       | 48,366,768,722     | 8.7               | 0.2%              | \$ 3,108           |
| <b>WALA (Year)</b> | Loan Count | Avg Balance | Count Curtailment | Amount Curtailment | % Curtailment     | % Bal Curtailment | Avg Curtail Amount |
| 1                  | 15,664,852 | 176,093     | 15,750,700        | 6,676,899,030      | 21.0%             | 0.242%            | \$ 2,029           |
| 2                  | 14,194,272 | 169,632     | 31,843,233        | 11,970,215,809     | 21.6%             | 0.497%            | \$ 3,906           |
| 3                  | 10,402,492 | 162,021     | 24,164,141        | 8,782,407,996      | 22.0%             | 0.521%            | \$ 3,831           |
| 4                  | 7,883,737  | 154,810     | 18,413,614        | 6,222,090,684      | 22.3%             | 0.510%            | \$ 3,541           |
| 5                  | 5,844,150  | 147,804     | 13,718,693        | 4,473,243,048      | 22.4%             | 0.518%            | \$ 3,414           |
| 6                  | 4,252,594  | 139,520     | 10,197,011        | 3,230,010,612      | 22.5%             | 0.544%            | \$ 3,382           |
| 7                  | 3,263,579  | 132,014     | 7,714,931         | 2,520,913,937      | 22.6%             | 0.585%            | \$ 3,419           |
| 8                  | 2,380,627  | 122,951     | 5,523,412         | 1,830,669,331      | 22.7%             | 0.625%            | \$ 3,395           |
| 9                  | 1,671,904  | 113,144     | 3,778,787         | 1,229,958,493      | 22.7%             | 0.650%            | \$ 3,236           |
| 10                 | 1,069,439  | 102,567     | 2,372,588         | 768,831,834        | 22.7%             | 0.701%            | \$ 3,170           |
| 11                 | 647,065    | 92,112      | 1,303,343         | 399,174,393        | 22.8%             | 0.670%            | \$ 2,707           |
| 12                 | 275,059    | 77,348      | 516,412           | 145,361,073        | 22.6%             | 0.683%            | \$ 2,344           |
| 13                 | 126,231    | 69,468      | 235,275           | 64,614,703         | 22.5%             | 0.737%            | \$ 2,274           |
| 14                 | 53,508     | 64,271      | 118,193           | 33,007,555         | 22.7%             | 0.960%            | \$ 2,722           |
| 15                 | 34,007     | 61,761      | 66,233            | 19,120,571         | 22.7%             | 0.910%            | \$ 2,479           |
| 16                 | 4,452      | 59,866      | 1,016             | 249,652            | 22.8%             | 0.094%            | \$ 246             |

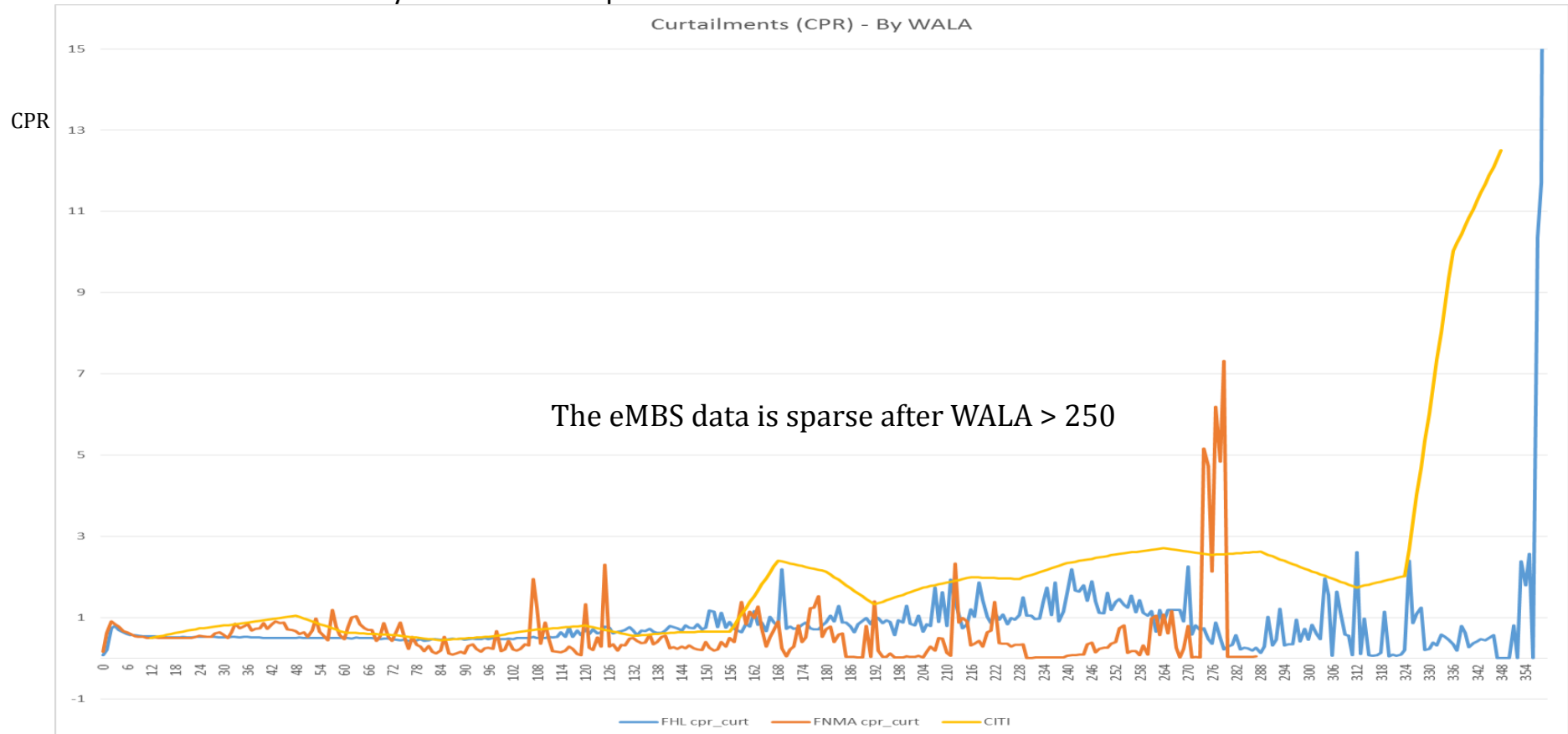
## Empirical Observations II

- Calculating Curtailments (GSE, eMBS):
  - Using a matching algorithm we pulled 6.5 million loans from the GSE and eMBS loan level data tables
  - Using different calculation methods we demonstrate that the observed curtailments are comparable between the two sources



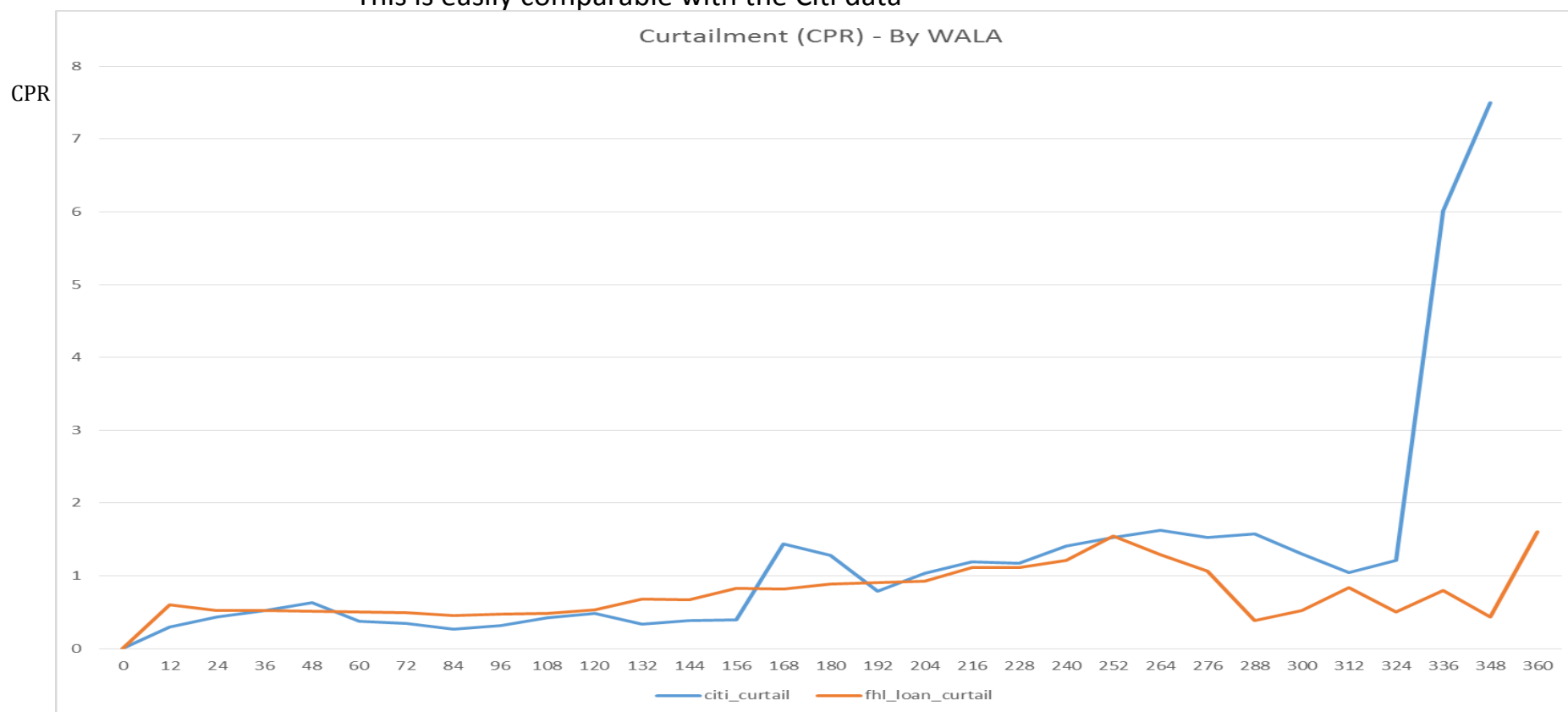
## Empirical Observations III

- Observations from the Data (Citi, GSE LL, eMBS):
  - Curtailments are very low and stable for the first 12-15 years of the life of the loans
  - The next 10-15 years are still stable, but curtailments become more significant @1-2 CPR
  - The last 3 years sees a sharp increase in curtailments as the borrowers seek to eliminate debt



## Empirical Observations IV

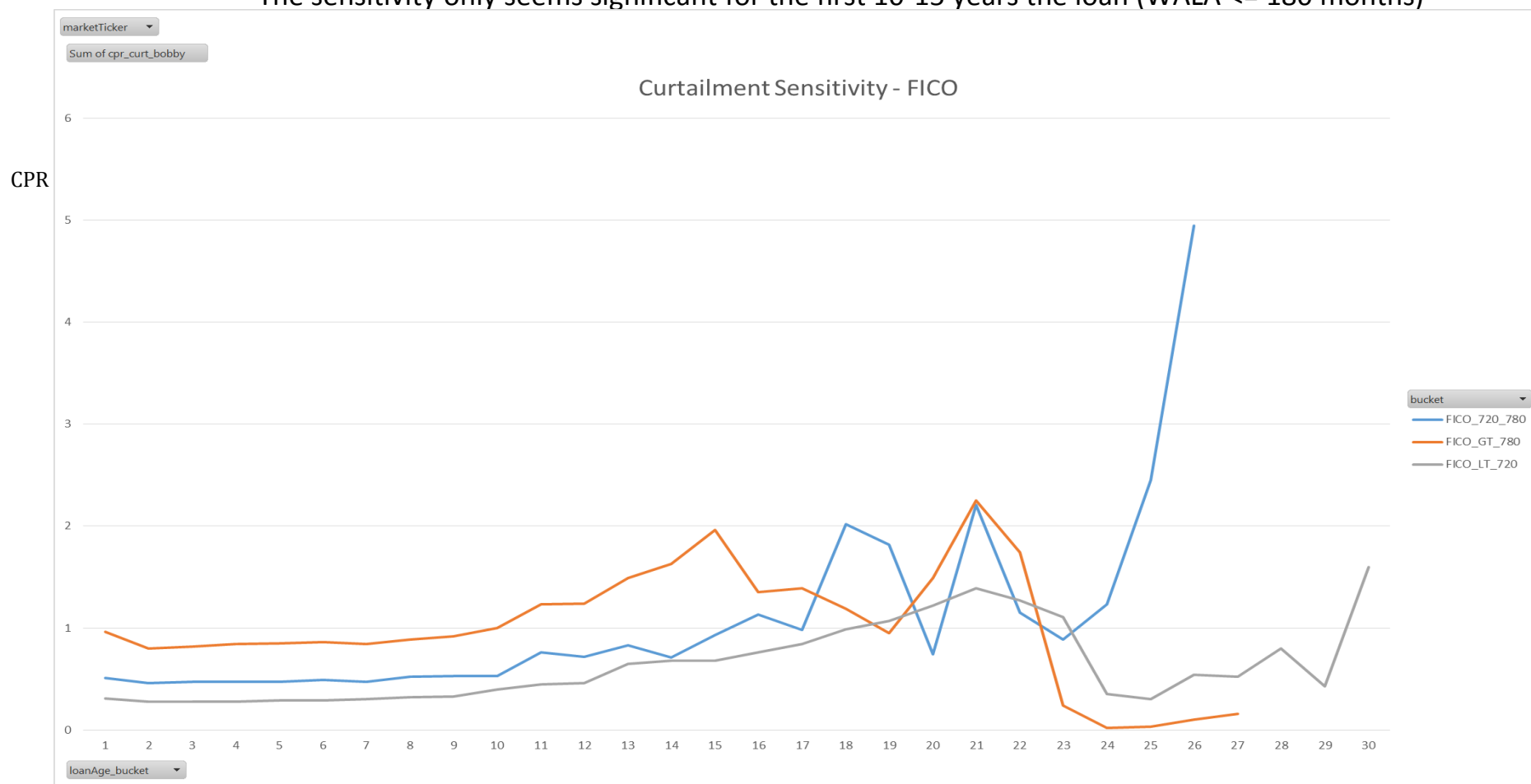
- FHLMC eMBS data vs Citi data
  - Freddie data has loan ages up to 360 months, so we use that data to calculate curtailments
  - The data is very sparse for the high WALA loans
    - Hence we group the data into Yearly Loan Age buckets
    - This is easily comparable with the Citi data



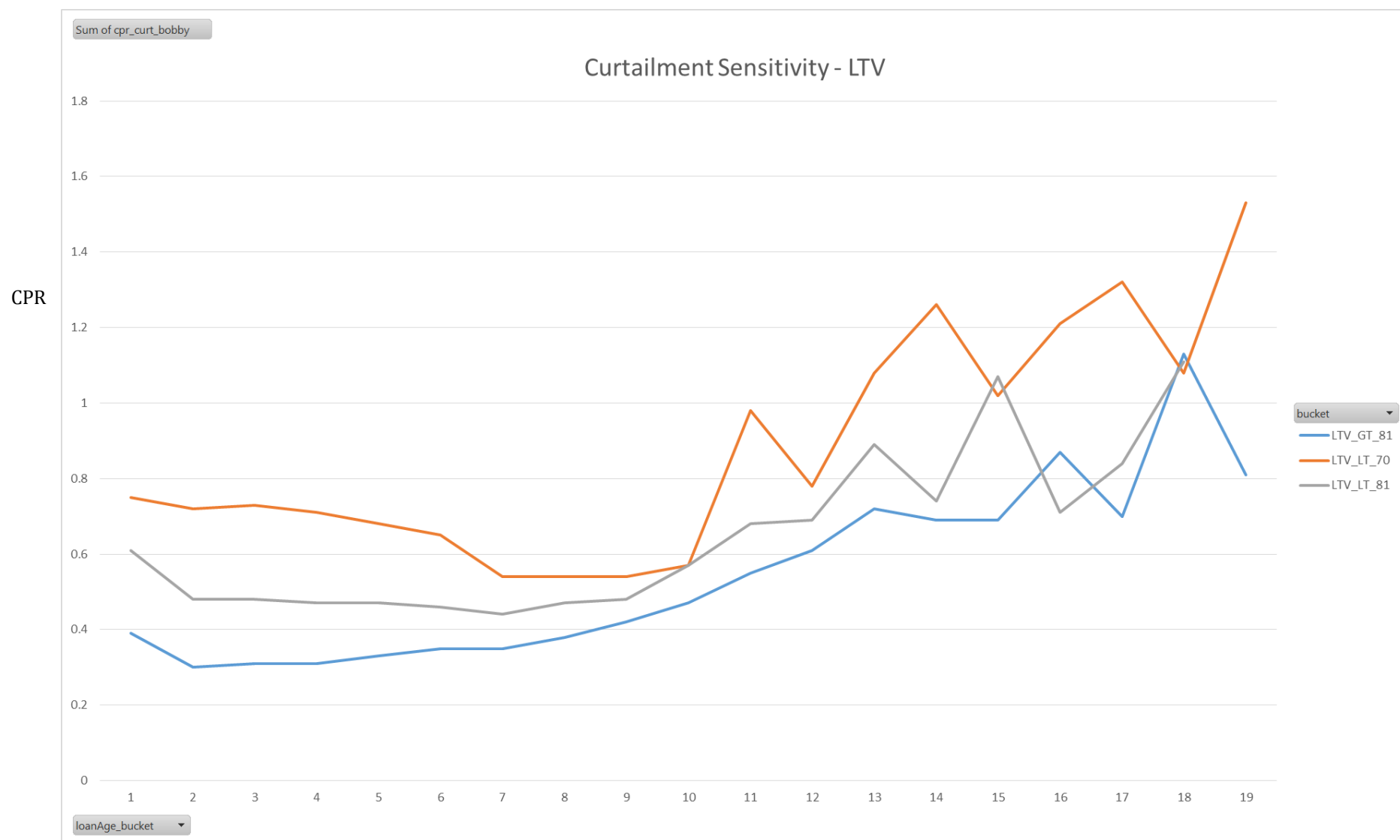


## Empirical Observations V

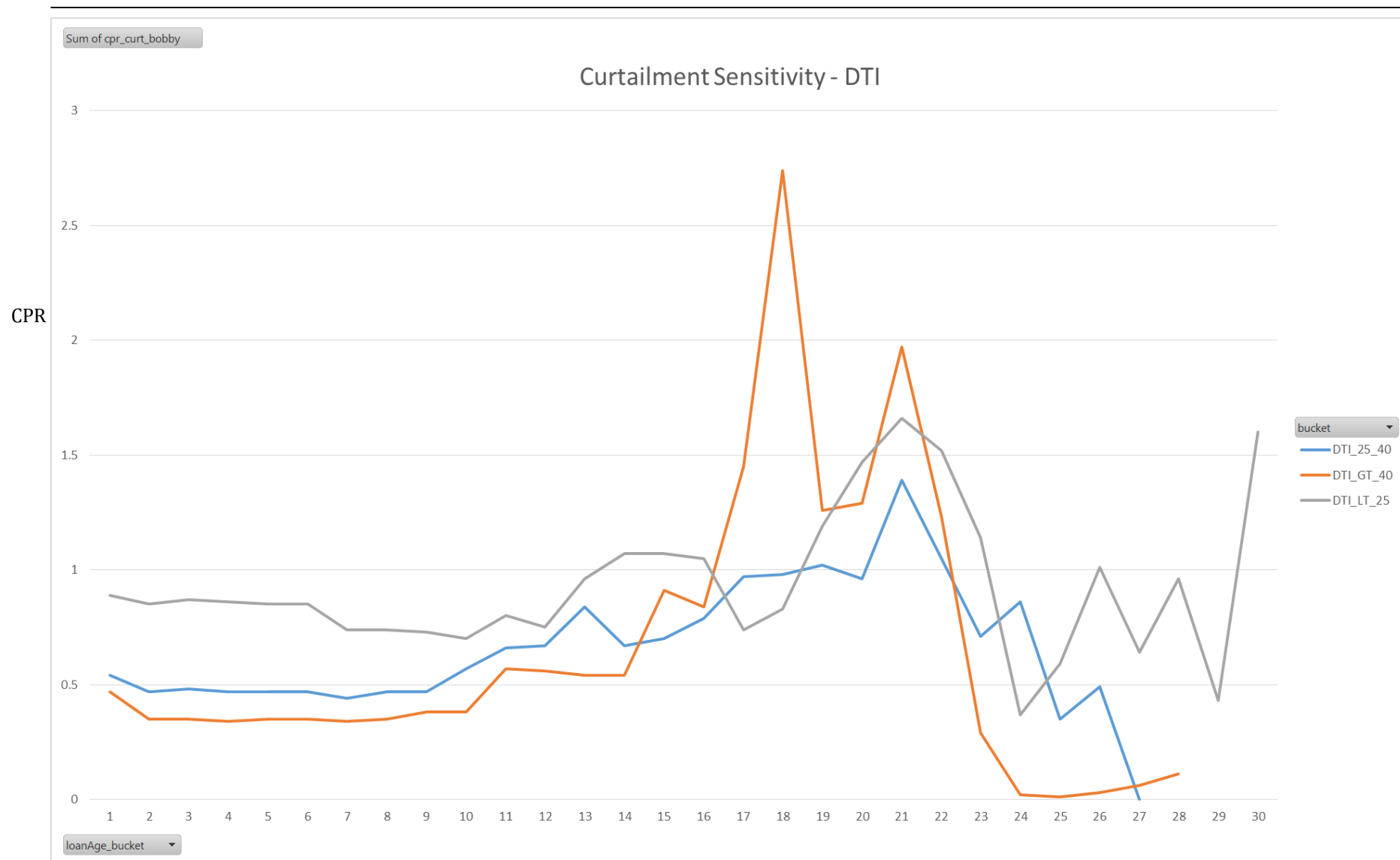
- Sensitivity of Curtailment to Collateral Attributes
  - Curtailments show dependence on FICO and LTV, DTI and Loan Size effects are less visible
  - The sensitivity only seems significant for the first 10-15 years the loan (WALA <= 180 months)



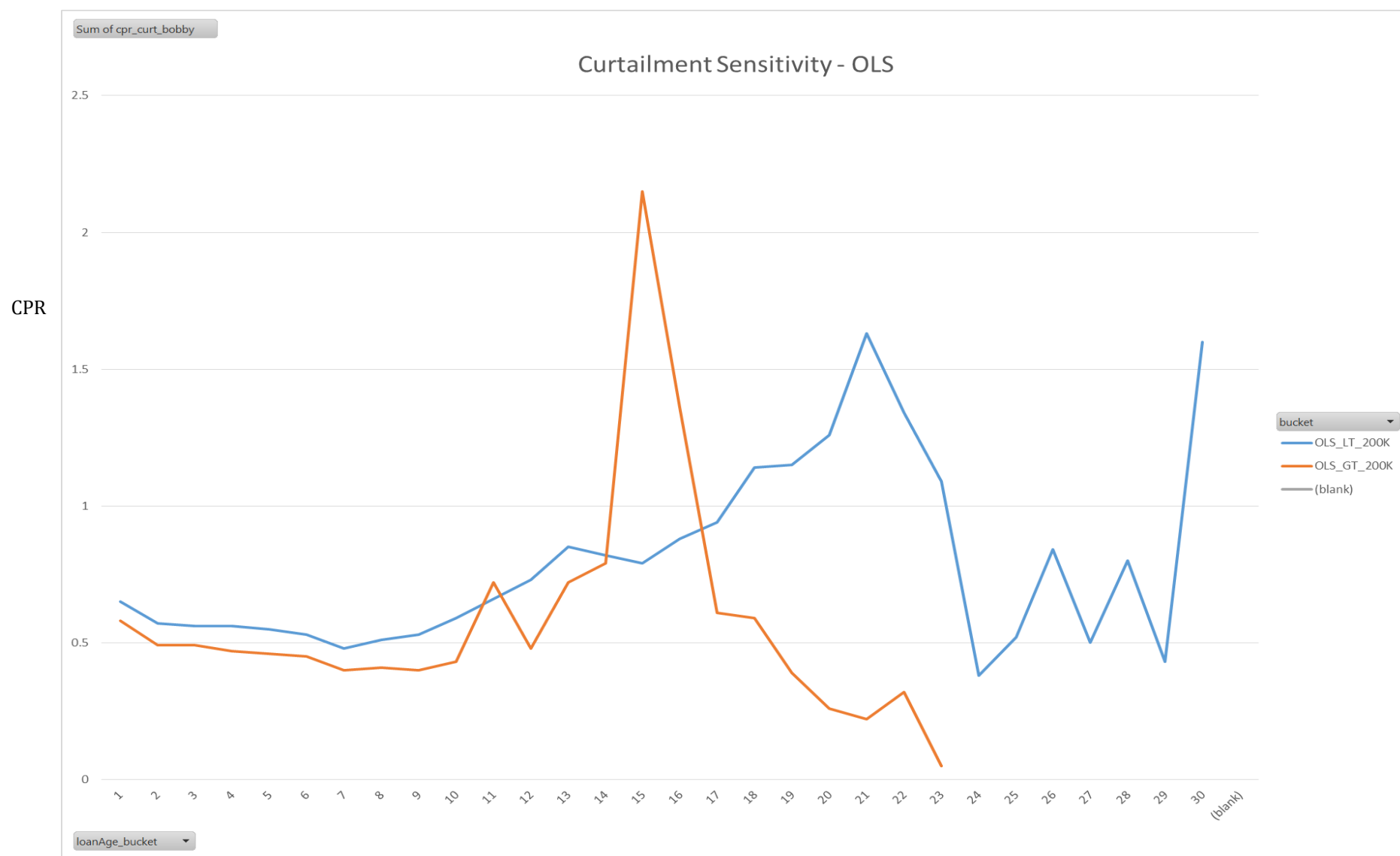
## Empirical Observations VI



## Empirical Observations VII

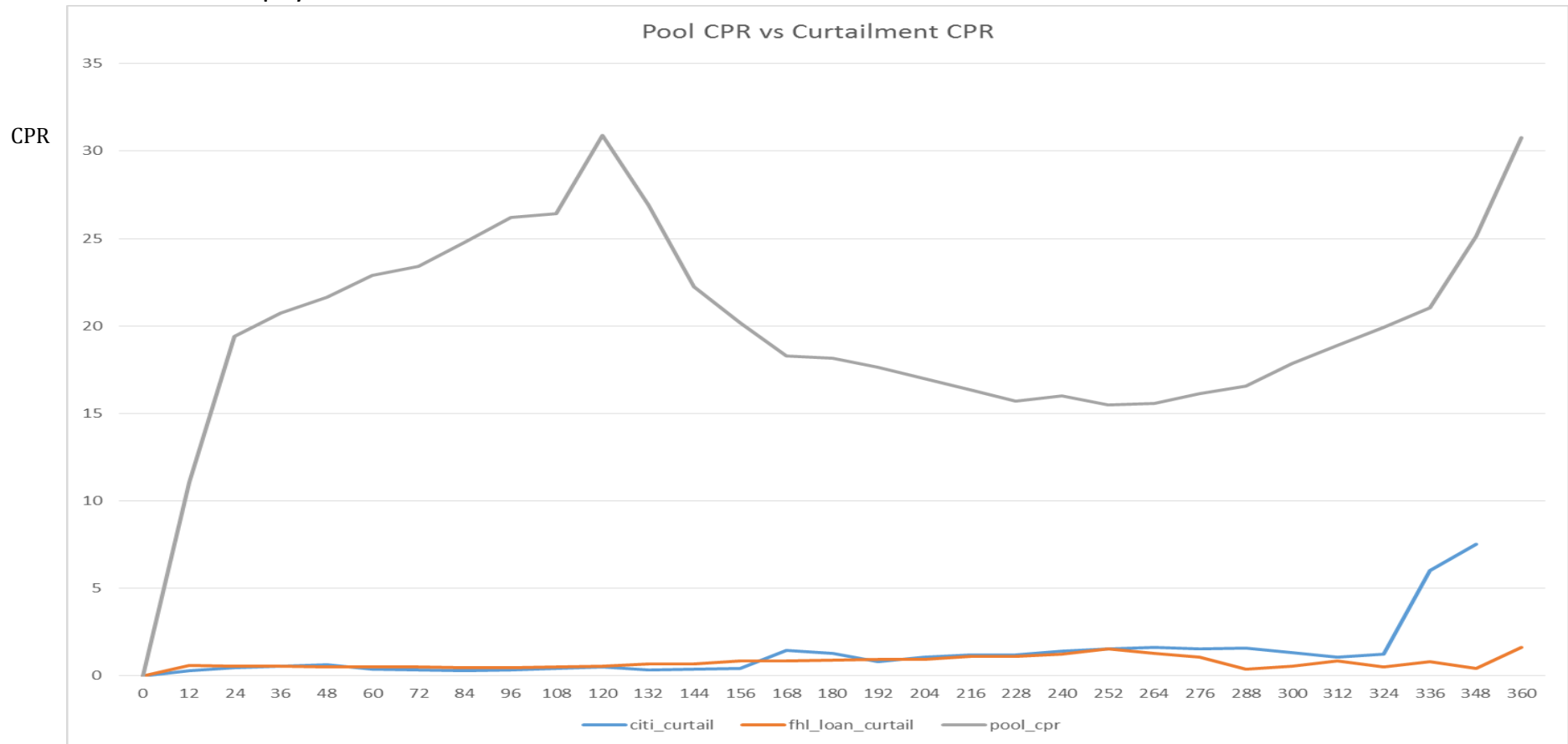


## Empirical Observations VIII



## Empirical Observations IX

- Final Data
  - We make use of the Citi Curtailment data along with the FHLMC Curtailment data
  - Additionally, we pull in the combined FHLMC/FNMA pool level CPR to help capture the full payoffs that we miss from our restrictive definition of curtailment



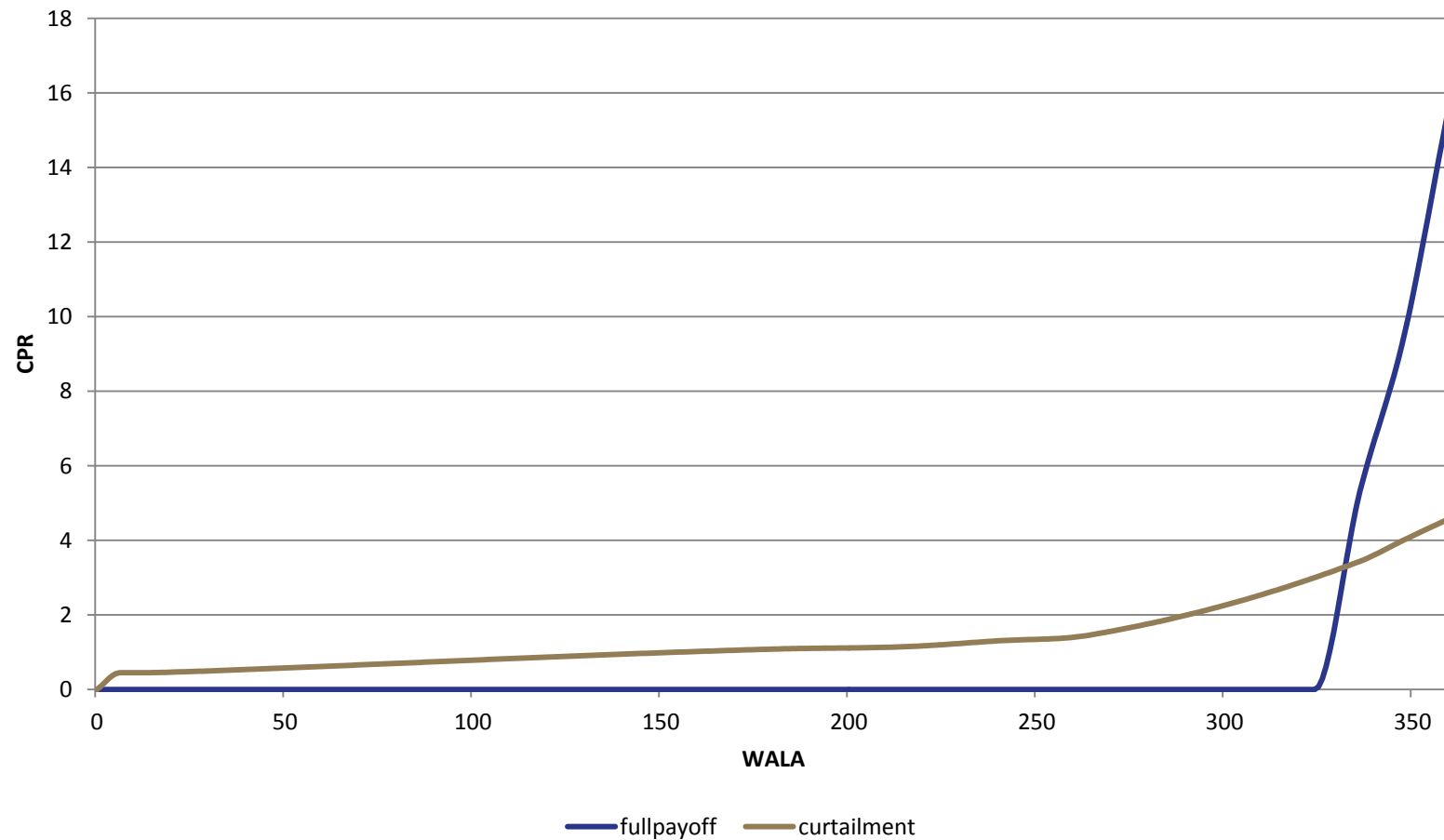
## Description of Framework

---

- We use R to combine and smooth the curtailment and full payoff data just described into a finalized curtailment vector for fixed-rate 30-year mortgages
- Curtailment is assumed to be a monotonically increasing function of WALA
  - It also shows some sensitivity to FICO and CLTV for the first 15 years. The dependence gradually decays to nothing over the 15 year period
- Care is taken to choose monotonically increasing data points to facilitate this assumption
- We make use of monotonically increasing spline functions to provide us with a smoothed fit to the data

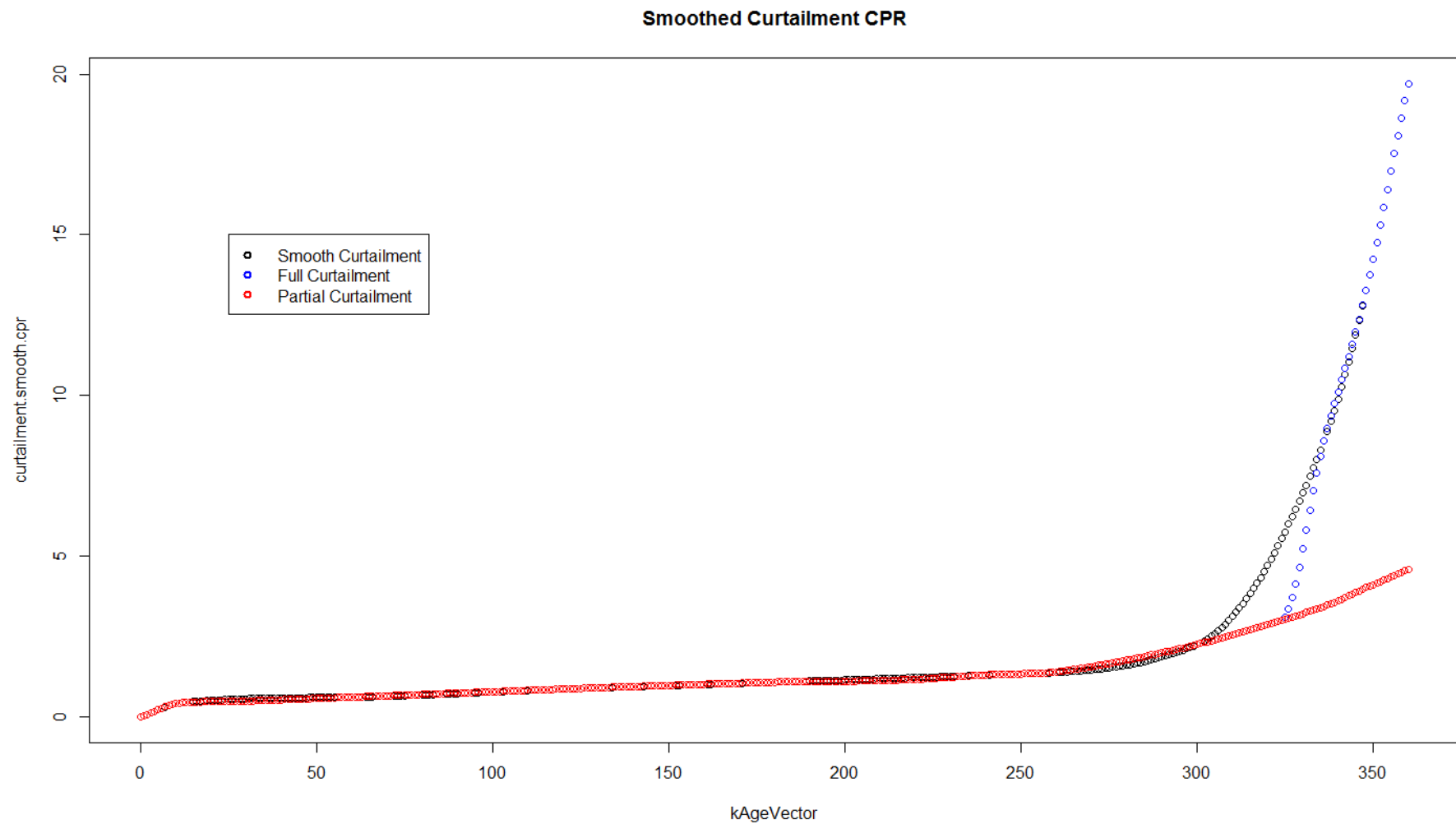
## Results I

- Curtailment and Full Payoff curves



## Results II

- Smoothed Curtailment and Full Payoff (Full Curtailment = Full Payoff)

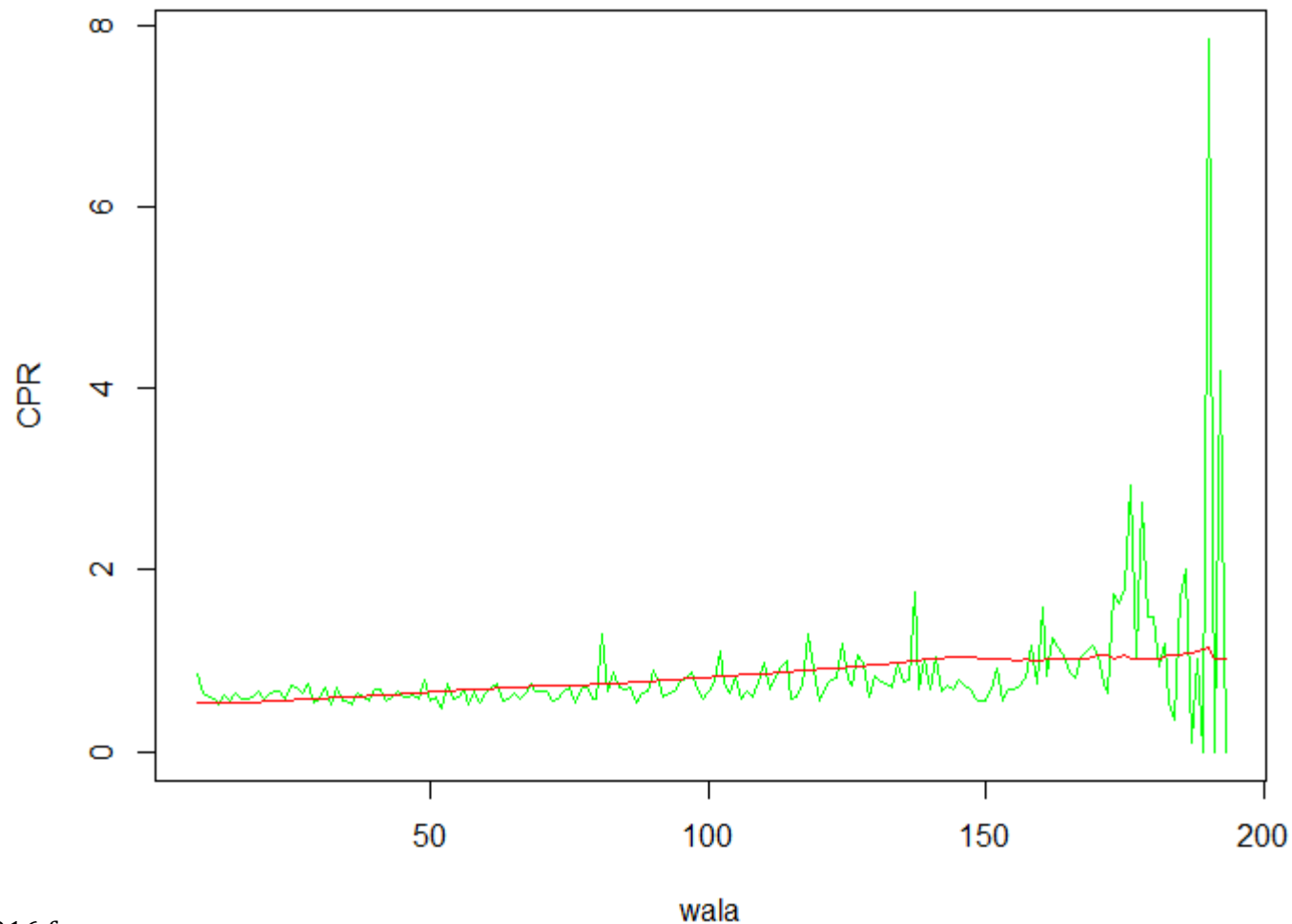




## Results III

---

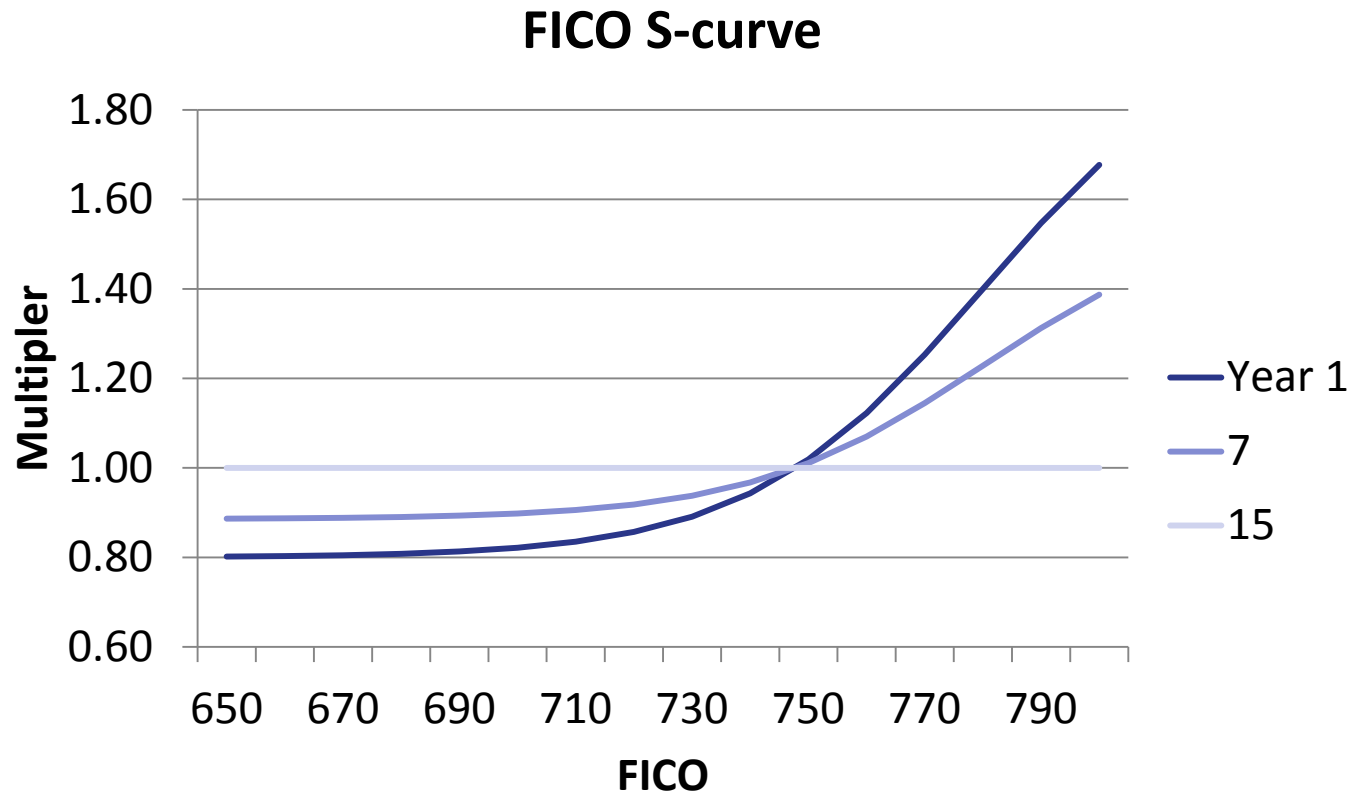
- For the GSE loan level data set, our model shows a close result compare with the actual curtailment numbers.



As of February 2016 factors

## Decay Factors

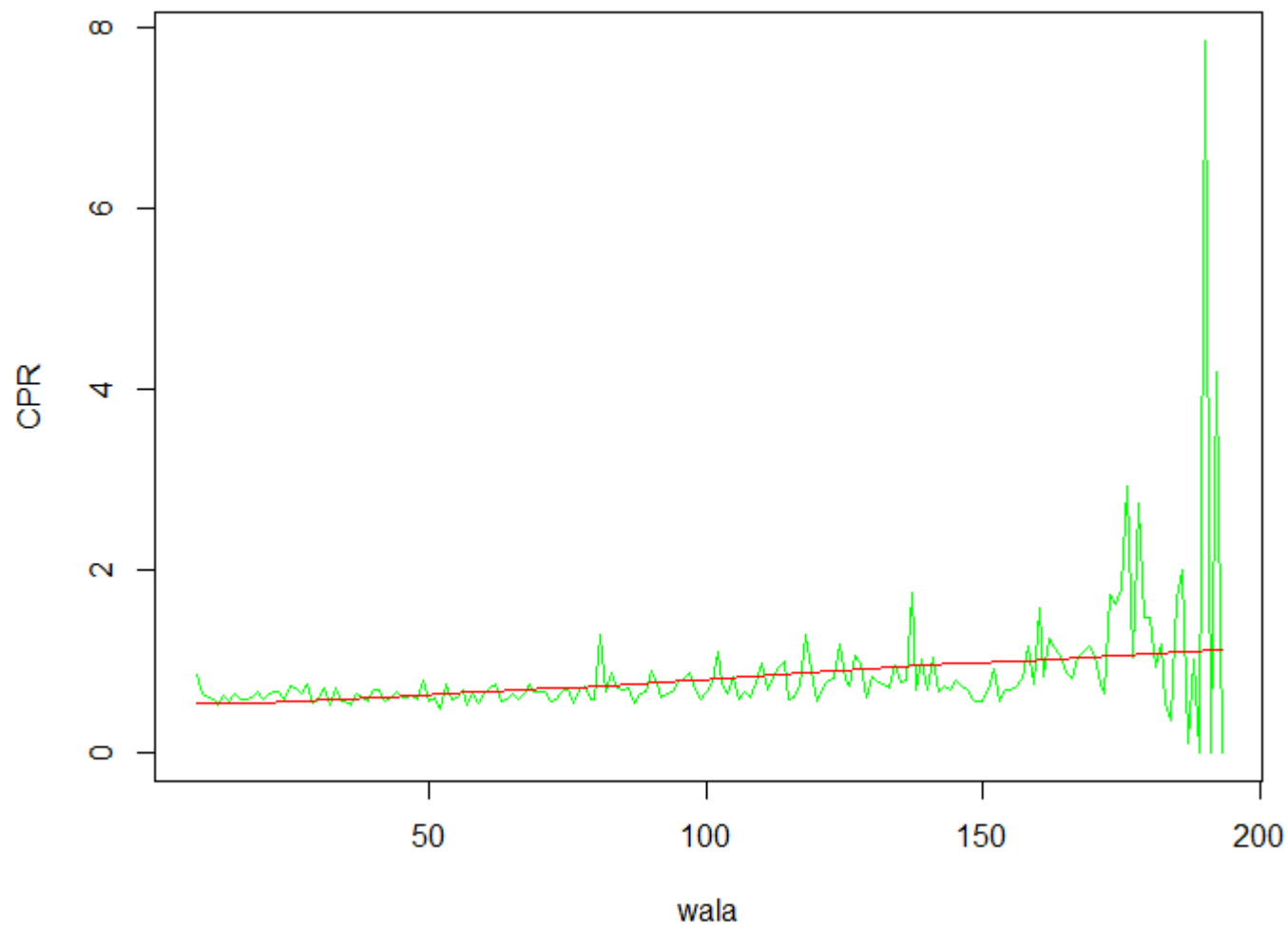
- The impacts of FICO and CLTV are not very clear for wala > 180 months. We allow FICO/CLTV multipliers to decay to 1 (zero effect) by 180 months.



## Tracking with Decay factors

---

- Introducing decay factors modestly improves the fits



## Total Tracking Error

- The combined version (FHL+FNM pools) shows our model is around 2% CPR slower before wala 125 months, 3.5% to 5% CPR faster than Actual after 125 months.

